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LETTER

# Forecasting short-term chlorophyll a concentration in Lake Erie using the machine learning XGBoost algorithm

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## Abstract

Harmful algal blooms are becoming increasingly prevalent due to climate warming and eutrophication. Leveraging machine learning tools to forecast algal blooms is crucial and promising for bloom management in various water systems. Notably, previous findings are site-specific, especially regarding the impacts of forecasting periods and important input features. However, there is a significant research gap in the application of machine learning for predicting algal blooms in the Great Lakes, the world's largest freshwater system. Thus, based on the measurements of 16 water quality parameters from 2012 to 2022, the author established the extreme gradient boosting (XGBoost) model to forecast chlorophyll *a* (Chl a, a proxy for algal biomass) concentration for 1–7 d in Lake Erie. The XGBoost model performance is quite satisfactory, with the lowest MSE of 10.94 and the highest $R^2$ of 0.99 for the 1 d forecast and an MSE of 83.90 and an $R^2$ of 0.90 for the 7 d forecast. Once trained, the model takes only a few seconds to run on an Intel Core i7 personal laptop. Based on Shapley additive explanations (SHAP) feature importance, water depth (Depth) and water temperature (Temp), are more important input features for the 7 d forecasting model than the well-recognized phosphorus and nitrogen nutrients, including particulate organic nitrogen (PON), soluble reactive phosphorous (SRP), nitrate + nitrite (NN), and total phosphorous (TP). Achieving relatively high accuracy for the 7 d forecast, with an $R^2$ of 0.83 and an MSE of 144.40, is possible by using only the top six most important input features: initial Chl a, Depth, Temp, PON, SRP, and NN, based on SHAP feature importance results. These findings highlight the accuracy and efficiency of the developed XGBoost model to predict Chl a in the world's largest freshwater system. The model can enhance algal bloom monitoring efficiency through early detection and key predictive features, supporting an early warning system for timely interventions, while also informing policy decisions and optimizing resource allocation.

## 1. Introduction

Due to climate warming and intensified human activities, harmful algal blooms have become increasingly prevalent in water bodies worldwide (Glibert 2020, Li *et al* 2022, Song 2023). Numerical models play a significant role in predicting and managing algal blooms such as issuing drinking water quality alerts, recreational notices, and fishing advisories (Ralston and Moore 2020, Astuti and Govindaraju 2025). While traditional process-based mechanistic models can offer explicit physical explanations, they are often computationally expensive due to intricate boundary conditions, strict convergence requirements, and the challenging calibration of numerous model parameters (Song *et al* 2023, Huang and Zhang 2024, Astuti and Govindaraju 2025). With the rise of the artificial intelligence era, efficient and advanced machine learning models skilled at handling non-linear relationships between variables emerge as powerful tools for simulating algal blooms in diverse water bodies (Glibert 2024, Park *et al* 2024).

The use of machine learning models to forecast algal blooms has become increasingly prevalent, as it can offer timely and essential information for decision-making and operational practices (Kim *et al*

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2022, Xia *et al* 2024, Szewczyk *et al* 2025). Notably, findings from previous studies are not always consistent and tend to vary on a case-by-case basis, likely due to distinct hydrological and environmental conditions. For example, researchers have found that one-month lead forecasts of Chlorophyll *a* (Chl a, which is widely used as a common proxy for algal biomass) using the extra tree regressor could be more accurate than current-month predictions using the random forest regressor in Biscayne Bay, USA (Yan *et al* 2024). In contrast, it was reported that the performance of deep-learning long short-term memory (LSTM) models decreased as the forecasting period increased from 1 d to 3 d in Lake Tuesday, USA (Liu *et al* 2022). Therefore, there is no consensus on how well models predict Chl a or on how the forecasting period affects model performance, as these effects may be site-specific. In addition, the important input features for building these models in different water bodies vary significantly. For instance, features such as current-month Chl a, land use, total phosphorus, and nitrogen oxides were determined to be critical for model performance in Biscayne Bay, USA (Yan *et al* 2024). Nevertheless, in Lake Danchi, China, and in the Nakdong and Yeongsan Rivers, South Korea, water temperature emerged as the most important feature for forecasting cyanobacterial blooms (Wang *et al* 2024). In Lake Geneva in Western Europe, nutrients are the key resources for algal growth and bloom development, being 30%–40% more influential than water temperature (Derot *et al* 2020).

The Laurentian Great Lakes form the world's largest freshwater system, serving as vital water resources for both the USA and Canada while periodically experiencing widespread and harmful algal blooms in Lake Erie (Ai *et al* 2023, Paerl *et al* 2024). In general, fewer studies have simulated algal blooms using machine learning models in the Great Lakes compared to other lakes around the world. Moreover, previous studies have mainly focused on applying machine learning models to forecast the risk of algal blooms using satellite images of Lake Erie (Ananias *et al* 2022, Tewari *et al* 2022, Ai *et al* 2023, Gupta *et al* 2023), which is framed as a classification problem. For example, the genetic algorithm was used to predict seasonal variations in the algal bloom severity index in Lake Erie, and its performance improved when large-scale climate indices were incorporated instead of relying solely on chemical nutrients (Tewari *et al* 2022). By building four machine learning models and introducing a satellite-derived cyanobacterial index, researchers found that the coarse timescale (30–60 d) variables, such as solar radiation, govern the algal growth dynamics even at finer timescales (15 d) (Gupta *et al* 2023). While these satellite-derived indices are easy to define and generally enhance prediction performance due to the limited number of classification categories (response variables), they primarily provide a qualitative classification. As a result, they lack a quantitative regression of algal biomass and fail to accurately interpret bloom intensity. Satellite images can also be uncertain and are significantly affected by cloud cover, making them less reliable than field observation data. In addition, almost no studies have conducted a direct forecast of algal biomass or algal blooms in this region, which is a regression problem. Thus, significant research gaps remain in using machine learning models to directly and accurately forecast algal growth in the Great Lakes, particularly concerning model performance, key input features, influencing environmental factors, and practical applications. It is also worth noting that previous studies primarily focused on the impact of nutrients on algal blooms in this region, while hydrodynamic conditions have not received much attention (Martin *et al* 2021, Green *et al* 2023, Paerl *et al* 2024).

To address the identified research gaps, the author utilized the most recent and comprehensive water quality data (2012–2022) to train the extreme gradient boosting (XGBoost) model for predicting Chl a concentration in Lake Erie. Predicting Chl a can assist in monitoring harmful algal blooms, guide water quality management, and inform early warning systems. The objectives are to (1) investigate the impacts of different forecasting periods (1–7 d) on the XGBoost model performance, (2) quantify the importance of input features for the established model, and (3) determine how reduced input features affect model performance, aiming to balance prediction accuracy with the number of measured water quality parameters used as inputs. These results can facilitate the application of machine learning models to forecast algal blooms in the Great Lakes.

## 2. Materials and methods

### 2.1. Study area and data preparation

The study region herein is western Lake Erie where annual algal blooms tend to occur (figure 1) (Ai *et al* 2023). Compared to the central and eastern regions of Lake Erie, the water depth in western Lake Erie is shallower, with an average depth of around 7.6 m, which likely promotes algal blooms (Ai *et al* 2023, Boegehold *et al* 2023). During the bloom-prone period (mainly May to October) from 2012 to 2022, water quality parameters were measured at ten sampling stations in western Lake Erie by personnel from the National Oceanic and Atmospheric Administration (NOAA) Great Lakes Environmental Research Lab (GLERL) and the Cooperative Institute for Great Lakes Research (CIGLR) (figure 1) (Boegehold *et al* 2023).

There are 16 measured water quality parameters including chlorophyll a (Chl a), water temperature (Temp), dissolved oxygen (DO), ammonia (AMM), nitrate + nitrite (NN), total phosphorous (TP), total dissolved phosphorous (TDP), soluble

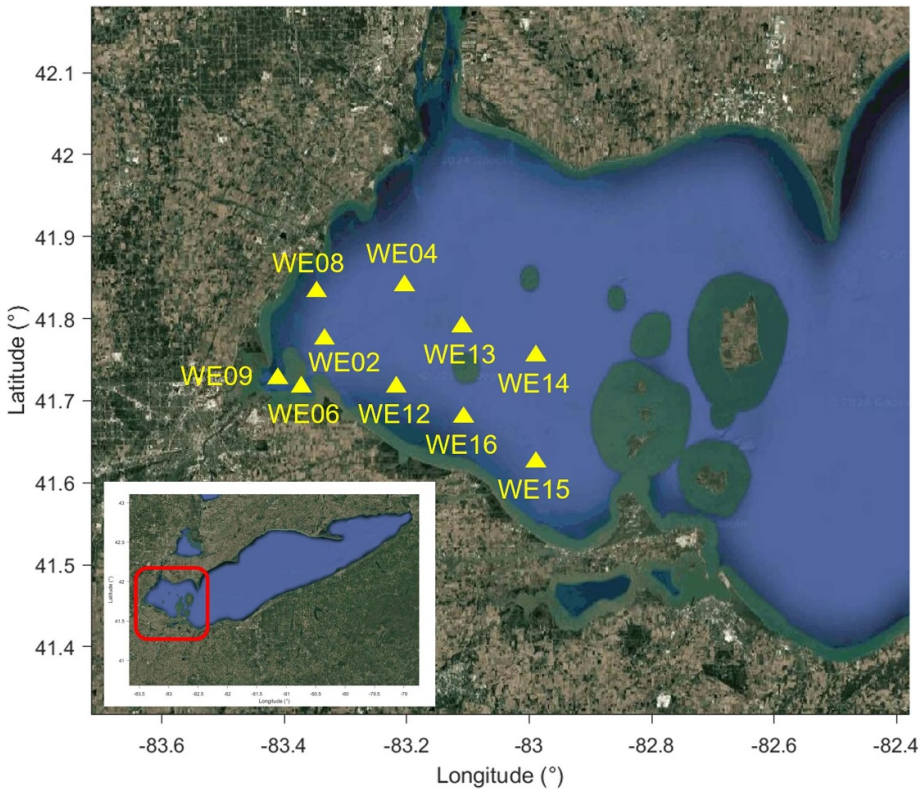



**Figure 1.** Locations of Lake Erie and water quality monitoring stations. The red box indicates the western Lake Erie region where algal blooms prevail. The yellow triangles indicate monitoring stations in the western Lake Erie.

reactive phosphorous (SRP), photosynthetically active radiation (PAR), Secchi depth (SD), total suspended solids (TSS), particulate organic carbon (POC), particulate organic nitrogen (PON), station depth (Depth), wind speed (WS), and conductivity (Cond) (Boegehold *et al* 2023) based on the NOAA National Centers for Environmental Information (NCEI) (source: https://doi.org/10.25921/11da-3x54) (Cooperative Institute for Great Lakes Research, University of Michigan; NOAA Great Lakes Environmental Research Laboratory 2019). The water parameters at ten monitoring stations were measured weekly, mostly between May and October, from 2012 to 2022, resulting in a dataset of 1420 samples. Moreover, distinct seasonal and interannual variations in Chl a concentration were detected, with the peak values observed in August over the 11 years (figure 2).

### 2.2. XGBoost machine learning model

#### *2.2.1. Model description*

The XGBoost machine learning model is a powerful ensemble tree-based regression model built on the gradient boosting framework (Chen and Guestrin 2016, Izadi *et al* 2021). Based on the author's preliminary tests (table S1), XGBoost was identified as the most suitable model in terms of accuracy among several machine learning models, including support vector machines, artificial neural networks (ANN), random forests (RF), etc. Additionally, tree-based models were found to be well-suited for predicting algal blooms in Lake Erie (Nguyen *et al* 2021, Song *et al* 2025). The XGBoost model is an optimized, scalable, and regularized implementation of gradient boosting decision trees (GBDT), incorporating core changes like second-order optimization, parallelized tree building, efficient handling of sparse data, and advanced regularization, making it highly efficient and robust (Sagi and Rokach 2021). In addition, tree-based models like XGBoost can be effectively interpreted using Shapley additive explanations (SHAP), which provide both local and global interpretability, enabling a deeper understanding of feature importance and interactions. Therefore, the XGBoost model was selected and first tested to forecast 1 d to 7 d Chl a concentration at all ten monitoring stations in this study. The XGBoost model is implemented in a Python 3.11 environment using the Scikit-learn package (version: 1.5.1) (Pedregosa *et al* 2011).

#### *2.2.2. Model setup*

To enhance the input data quality and model accuracy, an isolation forest method was used to remove outliers from the observed dataset (Cheng *et al* 2019). The contamination parameter in the isolation forest method was set to 0.05, indicating that the algorithm assumes 5% of the data points to be anomalies. Afterward, the K-nearest neighbors method was used to fill in the missing values, ensuring the dataset's integrity (Huang and Zhang 2024). The $K$ value of

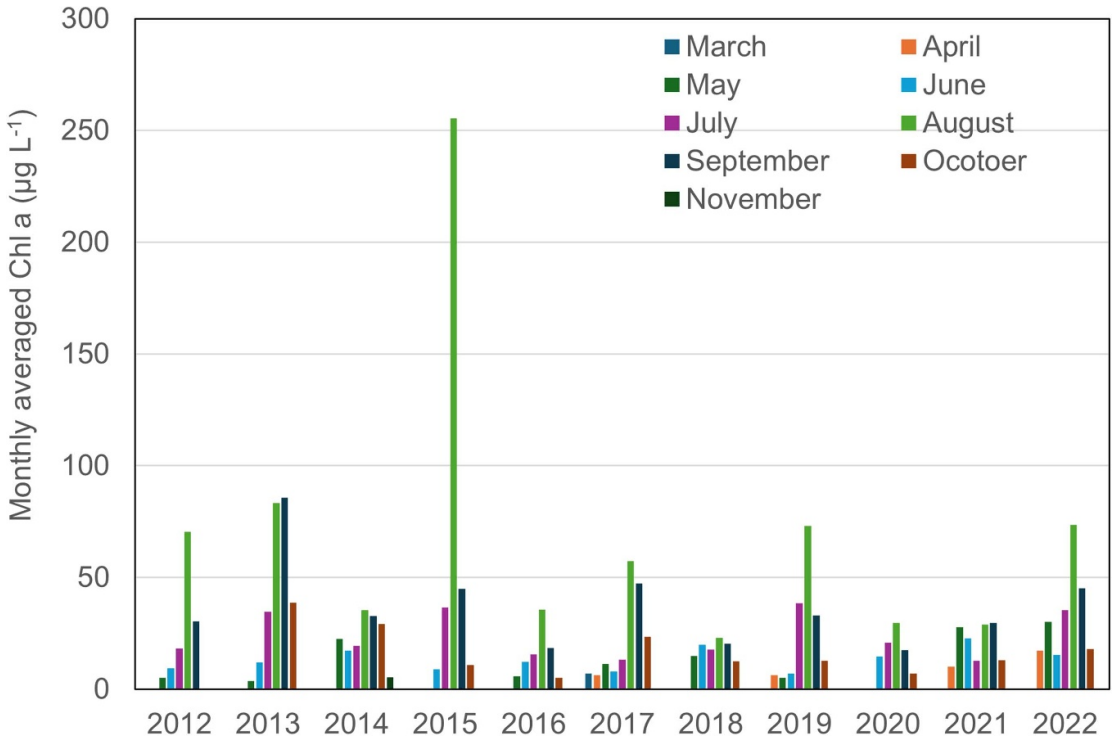



**Figure 2.** Monthly averaged Chl a concentration for ten monitoring stations in western Lake Erie.

the KNN imputation method is set to 5, meaning the algorithm uses the five nearest neighbors to estimate missing values. Then, the author applied linear interpolation to obtain the daily values from the weekly observed dataset. Note that interpolation was performed only within the intervals of the measured data. For example, in 2012, the weekly measurements were taken only from May to September. Therefore, interpolation for that year was limited to this period, without any extension beyond it. While PON and POC ($r = 0.98$) and SRP and TDP ($r = 0.93$) exhibit strong positive linear relationships (figure S1), the tree-based XGBoost model can remain fairly stable and robust in the presence of multicollinearity (Chen and Guestrin 2016, Chen *et al* 2020a, Guo *et al* 2021). Therefore, the four parameters above, which represent distinct aspects of water quality characteristics, were retained to comprehensively identify important parameters for model performance. Finally, standardization was applied to the input features to decrease the variability of input features and improve model robustness (Chen *et al* 2020b).

The XGBoost model outputs the future 1 d to 7 d Chl a, while the initial Chl a and the other 15 water quality parameters serve as input features. Please note that the initial Chl a was used only to forecast a single future Chl a value for individual dates (e.g. 4 d, 3 d, etc.), rather than using previous forecasts to predict later values. Therefore, the risk of using future data to predict the past is completely avoided. Then, the interpolated daily dataset was split into a training set (80%) and a testing set (20%). To optimize model performance, a grid search method combined with five-fold cross-validation is introduced to tweak hyperparameters within the model based on the training set (Izadi *et al* 2021, Yan *et al* 2024). The grid search for 7 d forecasting identified the optimal hyperparameters as follows: colsample_bytree (the fraction of features used to build each tree) = 1.0, learning_rate = 0.1, max_depth (the maximum depth of each decision tree) = 7, n_estimators (the number of trees) = 300, and subsample (the fraction of training data randomly sampled for building each tree) = 0.5 (table S2). The model performs quite well on the training set, with a mean squared error (MSE) of 3.58, a mean absolute error (MAE) of 1.37, and a coefficient of determination ($R^2$) of 1.00. Finally, the unseen testing set is used to reliably evaluate the model performance.

### 2.3. Model evaluation

In this study, MSE, MAE, and $R^2$ were used to evaluate model performance. The MSE is a convenient way to measure the average error and quantify the accuracy of predictions. MAE measures the average magnitude of errors between predicted and observed values. $R^2$ is commonly used to quantify regression model performance by measuring the proportion of the variance in the dependent variable that can be explained by the independent variable. The

mathematical equations of the three metrics are elaborated below.

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}-y)^2 \quad (1)$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}-y| \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(\hat{y}-y)^2}{\sum_{i=1}^{N}(\bar{y}-y)^2} \quad (3)$$

where $\hat{y}$ is the predicted value, $y$ is the observed value, $\bar{y}$ is the average of the observed values.

### 2.4. Analysis of feature importance

To better understand input features and enhance the interpretability of the XGBoost model, the SHAP method was used to quantify the importance of each input feature (Tao *et al* 2024). The SHAP method, based on cooperative game theory, assesses feature importance by measuring each input feature's marginal contribution across every possible subset of features and then averaging these contributions (Lundberg and Lee 2017). Thus, the method can ensure fair, consistent, and transparent feature attribution while maintaining additivity and being model-agnostic.

## 3. Results

### 3.1. Effects of different forecasting periods (1–7 d) on model performance

The XGBoost model performance in forecasting Chl a concentration for 1–7 d is basically acceptable with relatively small MSE values and high $R^2$ values (table 1), demonstrating the applicability of the machine learning model. Moreover, as the forecasting period increases, the model performance generally declines (table 1). The best model performance is observed in the scenario of the 1 d forecast with the lowest MSE of 10.94 and the highest $R^2$ of 0.99 (table 1). In contrast, the 7 d forecast scenario shows the worst prediction performance with the highest MSE of 83.90 and the lowest $R^2$ of 0.90 among all the forecasting period scenarios (figures 3 and S2). Interestingly, the model performance is moderate and relatively stable when the forecasting period is from 3 d to 6 d.

### 3.2. Input feature importance

The SHAP summary plot indicates that the initial Chl a, which has the largest SHAP value range, is the most influential feature affecting the performance of the XGBoost model in forecasting Chl a concentration specifically for the 7th day in the future (figure 4). Other key features influencing model performance include Depth, Temp, PON, and SRP, based on the larger distribution of the SHAP values (figure 4). High values of PON, SRP, and Temp (red points) increase the model prediction values (positive SHAP values), while high values of Depth (red points) decrease the predictions (negative SHAP values) (figure 4). Features like Wind and PAR show a more confined impact range and have relatively little impact on model output (figure 4). Moreover, the feature importance for forecasting Chl a over the 1 d to 6 d forecasting periods varies, with the initial Chl a and PON emerging as the consistently important features (figure S3).

**Table 1.** XGBoost model performance in predicting Chl a concentration with different forecasting periods.

| Forecasting period | Evaluation metrics | | |
|---|---|---|---|
| | MSE | MAE | $R^2$ |
| 1-day | 10.94 | 1.67 | 0.99 |
| 2-day | 35.11 | 2.96 | 0.96 |
| 3-day | 65.64 | 3.49 | 0.93 |
| 4-day | 69.64 | 4.11 | 0.93 |
| 5-day | 73.32 | 4.36 | 0.91 |
| 6-day | 63.63 | 4.26 | 0.92 |
| 7-day | 83.90 | 5.06 | 0.90 |

### 3.3. Model performance under different input feature combinations

Based on the feature importance analysis above, this study compared the XGBoost model performance in forecasting the 7 d Chl a concentration using different feature combinations (figure 5). Note that the order of parameter addition follows the feature importance obtained from the SHAP method (figure 5). Generally, adding more input water quality parameters can enhance the model performance with the decrement of MSE and the increment of $R^2$ (figure 5). Interestingly, the model performance reaches an $R^2$ of 0.83 and an MSE of 144.40 using only six input features: initial Chl a, Depth, Temp, PON, SRP, and NN (figure 5).

## 4. Discussion

The established XGBoost model demonstrates relatively strong performance in forecasting Chl a, with $R^2$ values ranging from 0.90 to 0.99 in Lake Erie (table 1), compared to previous studies where $R^2$ values ranged from 0.68 to 0.87 using RF, ANN, Bayesian, least absolute shrinkage and selection operator (LASSO), and LSTM (Feng Chang *et al* 2021, Ai *et al* 2023, Gupta *et al* 2023, Scavia *et al* 2023). While the data used and framework differ across the above studies, this comparison provides general information on model performance and highlights the strong potential of the XGBoost model for forecasting Chl a. The reason behind this could be the effectiveness and high performance of XGBoost, driven by second-order optimization and regularization to mitigate overfitting, along with the use of grid search for hyperparameter tuning (Chen and Guestrin 2016). Once trained, the

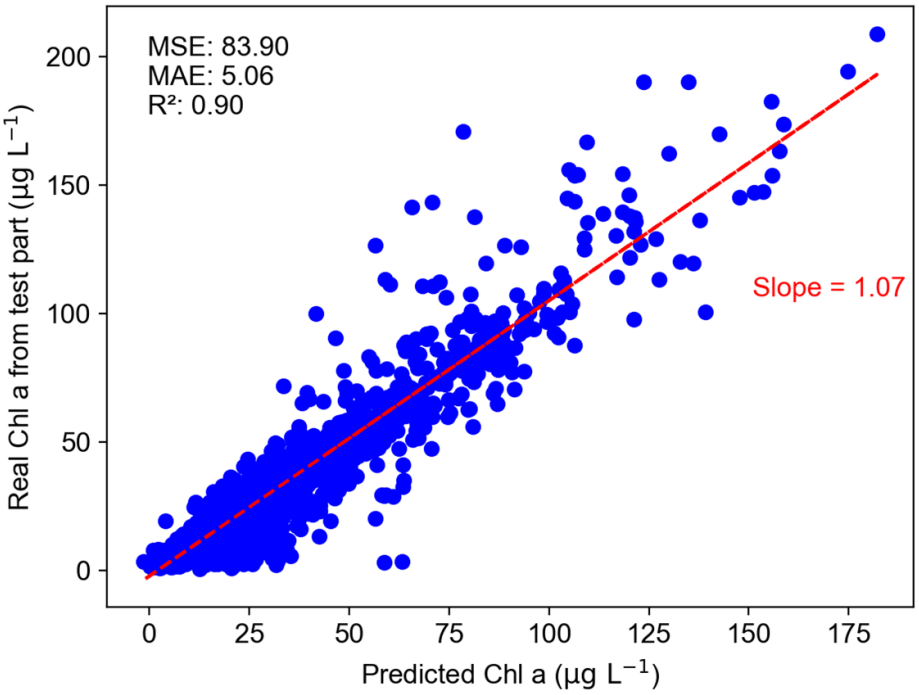



**Figure 3.** XGBoost model performance on the 7 d forecast of Chl a. The red line is the regression line. The slope value of the regression line close to 1 indicates high accuracy in the model predictions.

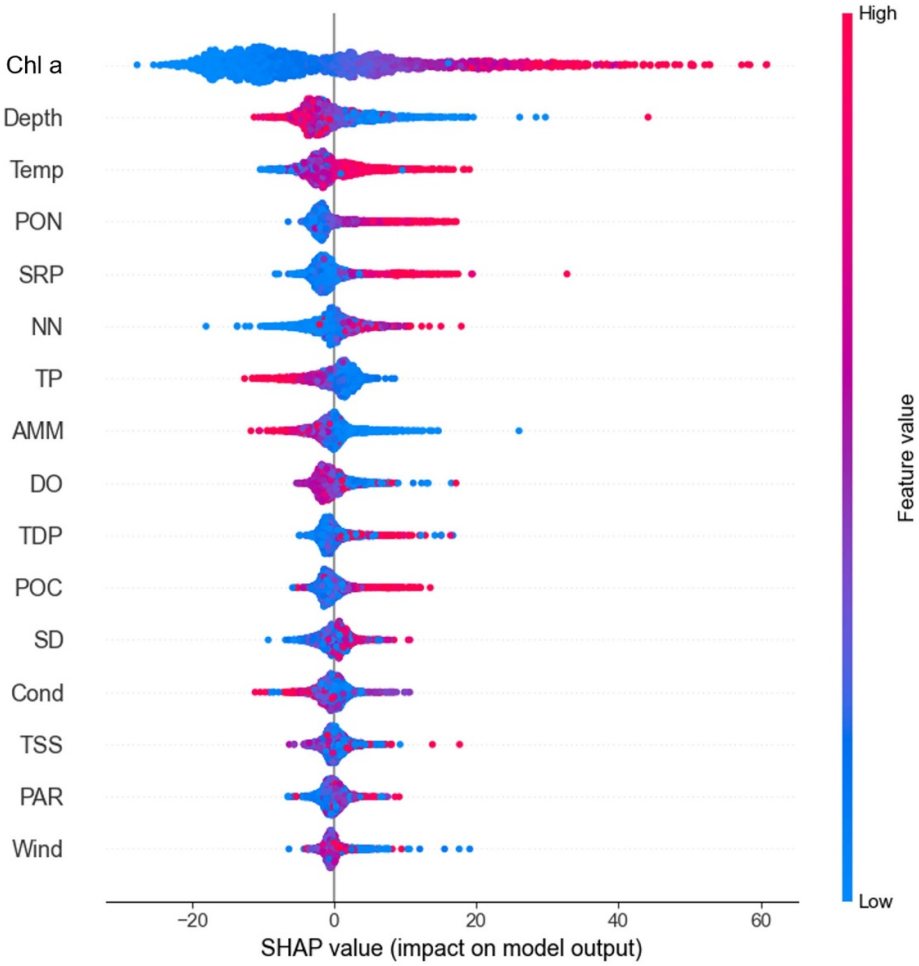



**Figure 4.** SHAP summary plot for the input features of the XGBoost model with the 7 d forecast of Chl a. The Chl a in the figure means the input initial Chl a.

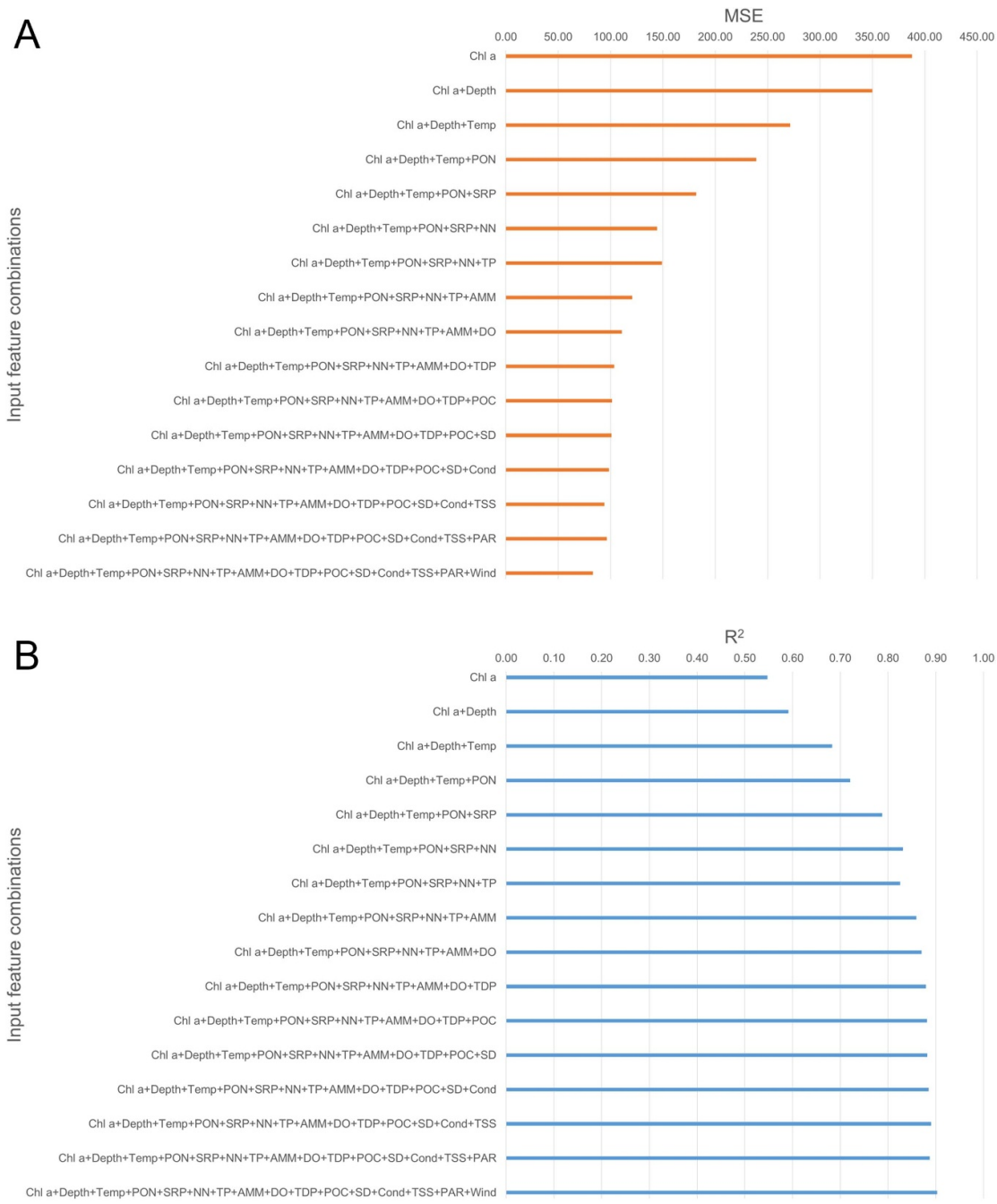



**Figure 5.** XGBoost model performance on the 7 d forecast of Chl a under different input feature combinations. The Chl a in the figure means the input initial Chl a.

XGBoost model runs in just a few seconds on a personal laptop equipped with an Intel Core i7-10510U (1.80GHz, 8 cores, boost up to ~2.3GHz) and 16GB of RAM. On the other hand, the XGBoost model performance decreases as the forecasting period increases from 1 d to 7 d, consistent with previous findings that increasing the forecasting period weakened model performance (Liu *et al* 2022, Gupta *et al* 2023). The reasons could be that as the prediction horizon increases, the relationship between the input features and the predicted variable weakens, older lagged features become less useful, and features that are relevant for short-term predictions lose importance (Hyndman and Athanasopoulos 2018). These results reveal the performance and applicability of the XGBoost model in forecasting Chl a in Lake Erie for the first time.

Based on the SHAP interpretation method, the author found that the input initial Chl a (the currently observed value) was the most important feature influencing the XGBoost model performance in forecasting the 1 d to 7 d Chl a (figures 4 and S3). This finding is consistent with previous studies (Xia *et al* 2024, Yan *et al* 2024), underscoring the importance of the initial

condition of Chl a content for model performance. Moreover, the feature importance results vary across different forecasting periods, with PON identified as the second most important feature for the 3 d to 6 d Chl a forecast (figure S3). This is generally consistent with the author's recent study on real-time Chl a prediction, in which PON emerged as the most critical water quality parameter (Song *et al* 2025). However, it is important to note that, based on the 7 d Chl a forecast, water depth and water temperature are more influential than PON. These differences highlight that forecasting and real-time prediction may rely on distinct key input features, suggesting the need for tailored approaches in each case.

Particularly, Depth and Temp were found more influential than PON, SRP, NN, and TP in forecasting the 7 d Chl a (figure 4), suggesting that hydrodynamic conditions can significantly influence Chl a concentration and algal bloom occurrence in western Lake Erie (Song *et al* 2021, Song 2023). For example, shallower water depth could facilitate sediment resuspension and nutrient release to the upper water (Dusini *et al* 2009, Shin *et al* 2023). Stable stratification and higher surface water temperatures favor algal blooms (Glibert 2020, Ishikawa *et al* 2022, Song *et al* 2023), especially the dominant blue–green algae, cyanobacteria, in Lake Erie (Yancey *et al* 2023). However, these hydrodynamic influences have been largely underestimated compared to the previous emphasis on nutrients (Scavia *et al* 2023, Kreiling *et al* 2024, Paerl *et al* 2024). Thus, future studies should devote more attention to the hydrodynamic factors (e.g., including flow velocity and water turbulence measurements) influencing algal blooms, rather than focusing solely on nutrient processes, to better understand and manage these events. A possible reason why Wind and PAR were the least important features is that light and wind conditions during the sampling periods (mainly May to October) are generally favorable for algal growth.

The impacts of input feature combinations on the XGBoost model performance were investigated herein. Surprisingly, only using 6 out of 16 measured water quality parameters, including initial Chl a, Depth, Temp, PON, SRP, and NN, can produce a relatively accurate model for the 7 d Chl a forecast with an $R^2$ of 0.83 and an MSE of 144.40 (figure 5). This finding highlights the essential roles of both hydrodynamic conditions and nutrients on predicting Chl a content and algal growth, consistent with previous results (Song *et al* 2018, Ishikawa *et al* 2022, Shin *et al* 2023, Song 2023). In addition, the reduced input feature combination could make future observations in Lake Erie more targeted and effective.

## 5. Conclusions

To promote algal bloom forecasting and management in the Great Lakes, this study established the XGBoost machine learning model to forecast Chl a concentration in Lake Erie based on the observed water quality data from 2012 to 2022. The main findings are presented as follows.

(1) The established XGBoost model exhibits strong performance in forecasting Chl a in Lake Erie, achieving $R^2$ values between 0.90 and 0.99 and MSE values between 83.90 and 10.94, surpassing previous studies.
(2) The model performance could deteriorate as the forecasting period increases, with the $R^2$ value of the 1 d forecast at 0.99 and that of the 7 d forecast at 0.90.
(3) Water depth and water temperature were found to have a greater impact than PON, SRP, and NN on the 7 d Chl a forecast, indicating that hydrodynamic conditions play a significant role in influencing Chl a concentration and the occurrence of algal blooms in Lake Erie. Future studies should place more emphasis on hydrodynamic mechanisms rather than solely relying on nutrient processes.
(4) Using a reduced set of input features, including initial Chl a, Depth, Temp, PON, SRP, and NN, can achieve relatively high accuracy for a 7 d forecast, with an $R^2$ of 0.83 and an MSE of 144.40.

These findings provide valuable insights for applying machine learning models to predict Chl a in the world's largest freshwater system, enhancing prediction accuracy and uncovering underlying mechanisms. The developed XGBoost model could improve algal bloom monitoring efficiency through early detection and by identifying key predictive features, supporting an early warning system for timely interventions. Using a reduced set of input features could improve the focus and effectiveness of water quality monitoring while reducing costs and maintaining a certain level of prediction accuracy for Chl a. These advancements can inform policy decisions, optimize resource allocation, and refine water quality regulations, contributing to more effective and cost-efficient algal bloom management strategies. However, due to the limitation of the sampling period, the XGBoost model was built based on the bloom-prone period (mainly from May to October). Future studies could focus on sampling more data in other months to further test model performance and refine it. Moreover, collecting data on flow velocity and water turbulence is also encouraged to enhance the model performance evaluation and mechanism explanation.

## Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.25921/11da-3x54.

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